

Topic I: Binary Search Trees – Invariants & Structural Properties

BST Property, Inorder Ordering, Min/Max Elements, Predecessors & Successors

Part 1: Benchmark Previous Year Questions (PYQs 1 to 3)

Q1. BST Structural Invariant Verification

GATE CS 2004 / DA Benchmark | MCQ

Let T be a binary tree where each node stores a distinct integer. Which of the following conditions is necessary and sufficient to confirm that T is a valid Binary Search Tree?

- (A) For every node x , $left(x) < x < right(x)$ (B) Inorder traversal of T yields a strictly ascending sequence
(C) Preorder traversal of T is sorted in ascending order (D) The minimum element is at the deepest left leaf

Answer: (B) Step-by-step Solution: While condition (A) only checks local parent-child relations (which fails if an indirect grandchild violates the global bound), an Inorder traversal (Left, Root, Right) produces a strictly sorted monotonic sequence if and only if all keys in the left subtree are smaller than the node, and all keys in the right subtree are larger. Thus, (B) is both necessary and sufficient.

Q2. Inorder Successor Location in a BST

GATE CS 2012 | MCQ

In a valid BST with distinct keys, if a node x has a non-empty right subtree, where is its inorder successor located?

- (A) The parent of node x (B) The minimum key in the right subtree of x
(C) The maximum key in the left subtree of x (D) The rightmost leaf in the entire tree

Answer: (B) Step-by-step Solution: In BST inorder traversal (Left, Root, Right), after visiting node x , the traversal visits the right subtree. The very first node visited inside the right subtree is its minimum element, obtained by traversing left pointers from the right child until reaching a node with no left child.

Q3. BST Distinct Keys Combinatorics

GATE CS 2005 / DA Sample | NAT

How many structurally distinct Binary Search Trees can be formed using 4 distinct keys {10, 20, 30, 40}?

Answer: 14 Step-by-step Solution: The number of structurally distinct binary search trees with n distinct keys is given by the n -th Catalan number: $C_n = (1 / (n + 1)) \times (2n! / (n! n!))$. For $n = 4$: $C_4 = (1 / 5) \times (8! / (4! \times 4!)) = (1 / 5) \times 70 = 14$.

Topic II: BST Search, Insert & Delete Trace Patterns (PYQs 4 to 6)

Q4. BST Search Probe Key Sequence Verification

GATE CS 2001 / 2010 | MCQ

Suppose we are searching for the key **60** in a BST containing distinct integers between 1 and 1000. Which of the following sequences of keys **CANNOT** be the sequence of nodes examined?

- (A) 2, 252, 401, 398, 330, 344, 397, 60
(B) 924, 220, 911, 244, 898, 258, 362, 60
(C) 925, 202, 911, 240, 912, 245, 60
(D) 2, 399, 387, 219, 266, 382, 381, 60

Answer: (C) Step-by-step Solution: When searching for 60, every comparison establishes an upper or lower bound: at 925 (go left: upper bound 925), at 202 (go right: lower bound 202), at 911 (go left: upper bound 911). At 240, target $60 < 240$, so we go left; the upper bound shrinks to 240. The next examined node cannot be 912 because $912 > 240$, violating the BST invariant. Thus, (C) is invalid.

Q5. Deletion of Node with Two Children

GATE CS 2015 | MCQ

In a BST, node X has two non-empty child subtrees. If X is deleted using the standard hibbard deletion strategy replacing X with its inorder predecessor, the node physically removed from the bottom of the tree has at most:

- (A) 0 children (leaf node)
(B) 1 child (which must be a left child)
(C) 1 child (which must be a right child)
(D) 2 children

Answer: (B) Step-by-step Solution: The inorder predecessor of X is the maximum element in X 's left subtree. Being the maximum element in that subtree, it cannot possess a right child (otherwise that right child would be larger). Thus, the predecessor has either 0 children or at most 1 child, which must be a left child.

Q6. BST Insertion Trace and Tree Height

GATE CS 2008 | NAT

Keys 10, 1, 3, 5, 15, 12, 16 are inserted sequentially into an initially empty BST. If the height of a single-node tree is defined as 0, what is the height of the resulting BST?

Answer: 3 Step-by-step Solution: Level 0: 10.
Level 1: Left child 1, Right child 15.
Level 2: 3 is right child of 1; 12 is left child of 15, 16 is right child of 15.
Level 3: 5 is right child of 3.
The deepest node is 5 at depth 3. Height = 3.

Q7. Truth Table Row Evaluation for Implication

GATE CS 2003 / DA Benchmark | MCQ

Which row in the truth table of $(P \wedge \neg Q) \rightarrow (R \vee P)$ evaluates to **FALSE**?

- (A) $P = \text{True}, Q = \text{False}, R = \text{False}$ (B) $P = \text{False}, Q = \text{False}, R = \text{False}$
 (C) $P = \text{True}, Q = \text{True}, R = \text{False}$ (D) No row (the formula is a tautology)

Answer: (D) Step-by-step Solution: An implication $A \rightarrow B$ is False if and only if $A = \text{True}$ and $B = \text{False}$. Here, $\text{LHS} = (P \wedge \neg Q)$, which is True only when $P = \text{True}$ and $Q = \text{False}$. If $P = \text{True}$, then $\text{RHS} = (R \vee P) = (R \vee \text{True}) = \text{True}$. Thus, whenever LHS is True, RHS is guaranteed True. It is never False under any assignment (tautology).

Q8. Number of Boolean Truth Assignments

GATE CS 2016 | NAT

How many rows in the complete truth table of a propositional formula with 3 variables P, Q, R evaluate to True for the sentence: $(P \leftrightarrow Q) \wedge (Q \leftrightarrow R)$?

Answer: 2 Step-by-step Solution: The formula asserts that P, Q , and R must all share the same truth value. With 3 boolean variables, there are $2^3 = 8$ total assignments. All three variables are identical in exactly two cases: (T, T, T) and (F, F, F) . Thus, exactly 2 rows evaluate to True.

Q9. Functionally Complete Operator Sets

GATE CS 2014 | MSQ

Which of the following sets of propositional logic connectives is/are **functionally complete**? [Select all that apply]

- (A) $\{\neg, \wedge\}$ (B) $\{\neg, \vee\}$
 (C) $\{\neg, \rightarrow\}$ (D) $\{\wedge, \vee\}$

Answer: (A), (B), and (C) Step-by-step Solution: A set of connectives is functionally complete if it can express every boolean truth function. $\{\neg, \wedge\}$ and $\{\neg, \vee\}$ are complete by De Morgan's laws. $\{\neg, \rightarrow\}$ is complete because $P \vee Q \equiv \neg P \rightarrow Q$. $\{\wedge, \vee\}$ cannot express negation $\neg$; hence it is not complete.

Q10. Complete 3-Variable Truth Table Analysis

GATE CS 2009 / DA Standard | MCQ

Construct the truth table for $\phi = (P \vee Q) \rightarrow (Q \wedge R)$. How many rows in the 8-row truth table evaluate to **TRUE**?

P	Q	R	$P \vee Q$	$Q \wedge R$	ϕ	P	Q	R	$P \vee Q$	$Q \wedge R$	ϕ
T	T	T	T	T	T	F	T	T	T	T	T
T	T	F	T	F	F	F	T	F	T	F	F
T	F	T	T	F	F	F	F	T	F	F	T
T	F	F	T	F	F	F	F	F	F	F	T

- (A) 3 (B) 4
(C) 5 (D) 6

Answer: (B) Step-by-step Solution: An implication is True when antecedent is False or consequent is True. Antecedent $(P \vee Q)$ is False in 2 rows: (F, F, T) and (F, F, F) $\rightarrow \phi$ is True. Consequent $(Q \wedge R)$ is True in 2 rows: (T, T, T) and (F, T, T) $\rightarrow \phi$ is True. In the remaining 4 rows: antecedent is True and consequent is False, making ϕ False. Total True rows = 2 + 2 = 4.

Q11. Inorder Predecessor in BST After Sequential Insertions

New Practice | NAT

The sequence 50, 30, 70, 20, 40, 60, 80, 35 is inserted into an empty BST. What is the key value of the inorder predecessor of node 40?

Answer: 35 Step-by-step Solution: Inorder traversal of the keys in ascending order: 20, 30, 35, 40, 50, 60, 70, 80. The key immediately preceding 40 in this sorted order is 35. Alternatively, 40 has left child 35 with no right descendants; thus the maximum in 40's left subtree is 35.

Q12. Valid Search Sequences for Target in BST

New Practice | MSQ

Keys 1 to 50 reside in a BST. We search for key 25. Which of the following sequences can be valid search paths? [Select all that apply]

- (A) 40, 10, 35, 20, 30, 25 (B) 10, 45, 15, 30, 28, 25
(C) 42, 12, 38, 22, 18, 25 (D) 5, 48, 12, 36, 20, 25

Answer: (A), (B), and (D) Step-by-step Solution: Track [lower, upper] interval for target 25:
(A): [1, 50] $\rightarrow$ 40: [1, 39] $\rightarrow$ 10: [11, 39] $\rightarrow$ 35: [11, 34] $\rightarrow$ 20: [21, 34] $\rightarrow$ 30: [21, 29] $\rightarrow$ 25. Valid!
(B): 10: [11, 50] $\rightarrow$ 45: [11, 44] $\rightarrow$ 15: [16, 44] $\rightarrow$ 30: [16, 29] $\rightarrow$ 28: [16, 27] $\rightarrow$ 25. Valid!
(C): At 22, 25 > 22, so lower bound becomes 23. Next examined node is 18 < 23 (violates lower bound). Invalid!

Practice Set: BST Tracing & Operations (Practice 13 to 15)

Q13. Hibbard Deletion Inorder Successor Substitution

[New Practice | MCQ](#)

In a BST containing keys {15, 8, 25, 20, 30, 18, 22}, root is 15. If node **20** is deleted by replacing it with its inorder successor, which node takes its place, and what is its parent?

- (A) Node 22 replaces 20; parent of 22 is now 25
(B) Node 18 replaces 20; parent of 18 is now 15
(C) Node 25 replaces 20; parent of 25 is now 15
(D) Node 22 replaces 20; parent of 22 is now 18

Answer: (A) **Step-by-step Solution:** Node 20 has left child 18 and right child 22. Its inorder successor is the minimum node in its right subtree, which is 22. In hibbard deletion, key 22 is copied into node 20's position. The parent of this position is 25 (since 20 was the left child of 25).

Q14. Skewed BST Worst-Case Search and Insert Bounds

[New Practice | MSQ](#)

Keys 1, 2, 3, ..., n are inserted in strictly increasing sorted order into an initially empty BST. Which of the following statements is/are **TRUE**? [Select all that apply]

- (A) The resulting BST is a right-skewed tree of height $n - 1$.
(B) The worst-case search time complexity becomes $\Theta(n)$.
(C) The inorder traversal sequence remains 1, 2, 3, ..., n .
(D) Finding the minimum element takes $\Theta(n)$ time.

Answer: (A), (B), and (C) **Step-by-step Solution:** Inserting sorted keys produces a degenerated linked-list structure (right-skewed) of height $n - 1$, degrading search/insert to $\Theta(n)$. Inorder traversal on any valid BST always yields ascending order (C). (D) is false: the minimum element is at the root node, found in $O(1)$ time.

Q15. BST Range Search Comparisons Count

[New Practice | NAT](#)

Consider a BST with keys: 50 (root), left subtree {30, 20, 40}, right subtree {70, 60, 80}. A range query reports all keys in [35, 65]. How many node keys fall strictly inside this query interval?

Answer: 3 **Step-by-step Solution:** The keys present in the tree are {20, 30, 40, 50, 60, 70, 80}. Checking each key against $35 \leq k \leq 65$: 40 (yes), 50 (yes), 60 (yes). Keys 20, 30, 70, 80 are outside the range. Exactly 3 keys satisfy the condition.

Practice Set: Truth Table Construction & Normal Forms (Practice 16 to 18)

Q16. Truth Table Minterm Expansion

[New Practice | MCQ](#)

What is the sum of minterms ($\sum m$) representation of the boolean formula $(P \wedge Q) \vee (\neg P \wedge R)$ over variables (P, Q, R) where P is MSB?

- (A) $\sum m(1, 3, 6, 7)$ (B) $\sum m(2, 3, 6, 7)$
(C) $\sum m(1, 2, 5, 7)$ (D) $\sum m(0, 1, 6, 7)$

Answer: (A) Step-by-step Solution: • $(\neg P \wedge R)$ requires $P = 0, R = 1$. This matches $(P, Q, R) = (0, 0, 1)$ [minterm 1] and $(0, 1, 1)$ [minterm 3].

• $(P \wedge Q)$ requires $P = 1, Q = 1$. This matches $(P, Q, R) = (1, 1, 0)$ [minterm 6] and $(1, 1, 1)$ [minterm 7].

The union of satisfying assignments gives minterms $\{1, 3, 6, 7\}$.

Q17. Parity Function Truth Table Count

[New Practice | NAT](#)

How many rows in the 8-row truth table evaluate to True for the 3-variable XOR formula: $P \oplus Q \oplus R$?

Answer: 4 Step-by-step Solution: The XOR of three variables is True if and only if an odd number of variables are True (1 or 3 variables are True).

• Exactly one True: $(T, F, F), (F, T, F), (F, F, T) \rightarrow 3$ rows.

• All three True: $(T, T, T) \rightarrow 1$ row.

Total True rows = $3 + 1 = 4$ rows. (Even parity assignments produce False).

Q18. Equivalence via Truth Table Comparison

[New Practice | MSQ](#)

Which of the following formulas have identical truth tables to $\neg(P \rightarrow (Q \vee R))$? [Select all that apply]

- (A) $P \wedge \neg Q \wedge \neg R$ (B) $P \wedge \neg(Q \vee R)$
(C) $\neg P \vee Q \vee R$ (D) $(P \wedge \neg Q) \vee (P \wedge \neg R)$

Answer: (A) and (B) Step-by-step Solution: Recall $\neg(A \rightarrow B) \equiv A \wedge \neg B$. Letting $A = P$ and $B = Q \vee R$ gives: $P \wedge \neg(Q \vee R)$ (option B).

Applying De Morgan's law: $P \wedge \neg Q \wedge \neg R$ (option A). (C) is the negation of the target. (D) is equivalent to $P \wedge (\neg Q \vee \neg R) \neq P \wedge \neg Q \wedge \neg R$.

Q19. Complete Truth Table Construction: Majority Function

[New Practice | MCQ](#)

Consider the majority function formula: $\psi = (P \wedge Q) \vee (Q \wedge R) \vee (P \wedge R)$. Its truth table output column across standard ordering (TTT to FFF) is:

P	Q	R	$P \wedge Q$	$Q \wedge R$	$P \wedge R$	ψ
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	F	F	T	T
T	F	F	F	F	F	F
F	T	T	F	T	F	T
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

(A) [T, T, T, F, T, F, F, F]

(B) [T, T, F, F, T, F, F, F]

(C) [T, F, T, F, T, F, T, F]

(D) [T, T, T, T, F, F, F, F]

Answer: (A) Step-by-step Solution: ψ evaluates to True whenever at least 2 of the 3 variables are True. The rows with ≥ 2 Trues are: TTT, TTF, TFT, FTT (rows 0, 1, 2, 4 in binary index 7, 6, 5, 3). Rows with ≤ 1 True evaluate to False. The resulting output column is strictly [T, T, T, F, T, F, F, F].

Q20. Biconditional Chaining Truth Table

[New Practice | NAT](#)

In the truth table for the formula $((P \rightarrow Q) \wedge (Q \rightarrow R)) \leftrightarrow (P \rightarrow R)$, how many rows evaluate to **FALSE**?

Answer: 1 Step-by-step Solution: Let LHS = $(P \rightarrow Q) \wedge (Q \rightarrow R)$ and RHS = $(P \rightarrow R)$. By hypothetical syllogism, LHS $\rightarrow$ RHS is always True. For the biconditional $\leftrightarrow$ to be False, RHS must be True while LHS is False. Consider $(P = F, Q = T, R = F)$: RHS $(F \rightarrow F)$ is True. But in LHS: $Q \rightarrow R$ ($T \rightarrow F$) is False, making LHS False! Thus, under (F, T, F), LHS is False and RHS is True. All other 7 rows have matching truth values. Exactly 1 row evaluates to False.

Q21. Self-Dual Boolean Formulas via Truth Tables

[New Practice | MCQ](#)

A boolean function $f(P, Q, R)$ is *self-dual* if $f(\neg P, \neg Q, \neg R) = \neg f(P, Q, R)$. Which of the following functions is self-dual?

(A) Majority function: $(P \wedge Q) \vee (Q \wedge R) \vee (P \wedge R)$

(B) $P \wedge Q \wedge R$

(C) $P \vee Q \vee R$

(D) $(P \rightarrow Q) \wedge R$

Answer: (A) Step-by-step Solution: In the majority function, swapping all variables with their complements changes a state with k Trues into a state with $(3 - k)$ Trues. Since $k \geq 2 \Leftrightarrow 3 - k \leq 1$, the output is guaranteed to flip from True to False and vice-versa for every row. Hence, the majority function is self-dual.

Synthesis & Cross-Topic Mastery (Practice 22 to 24)

Q22. Number of Binary Trees vs Number of BSTs

[New Practice | NAT](#)

Given 3 labeled keys {1, 2, 3}. How many structurally distinct **Binary Search Trees** can be formed using all 3 keys?

Answer: 5 Step-by-step Solution: For a fixed set of distinct keys, each unique binary tree shape corresponds to exactly ONE valid BST assignment (inorder is uniquely determined). The number of binary tree shapes with 3 nodes is the 3rd Catalan number: $C_3 = (1/4) \times (6! / (3! 3!)) = (1/4) \times 20 = 5$.

Q23. Tautology Testing without Full Truth Table

[New Practice | MSQ](#)

Which of the following propositions can be proven to be tautologies by showing their falsifying assignment produces a contradiction? [Select all that apply]

(A) $((P \rightarrow Q) \wedge \neg Q) \rightarrow \neg P$ (Modus Tollens)

(B) $(P \wedge Q) \rightarrow P$

(C) $(P \vee Q) \rightarrow Q$

(D) $(P \rightarrow (Q \rightarrow R)) \rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$

Answer: (A), (B), and (D) Step-by-step Solution: (A) is Modus Tollens. (B) is Simplification. (D) is Frege's theorem (distribution of implication). All three are valid tautologies whose falsification attempts force simultaneous True and False assignments to the same atomic variable. (C) is falsified by setting $P = \text{True}$, $Q = \text{False}$ (not a tautology).

Q24. Maximum Key Searches in a BST of Height 4

[New Practice | NAT](#)

In a BST of height 4 (where a single root node has height 0), what is the maximum number of key comparisons required in an unsuccessful search?

Answer: 5 Step-by-step Solution: In a BST of height 4, the deepest leaf resides at depth 4. The path from the root to this deepest leaf contains: depth 0, 1, 2, 3, 4, which is exactly 5 nodes. An unsuccessful search that terminates at a null pointer below this leaf performs 1 comparison at each of the 5 nodes along the path. Total comparisons = 5.